- Most of the compounds of transition elements are paramagnetic. Magnetic properties are related to the electronic configuration of atoms.
- Many industrial processes use transition metals or their compounds as catalysts. Transition metal has energetically available d orbitals that can accept electrons from reactant molecule or metal can form bond with reactant molecule using its d electrons.
- Transition metals form a number of interstitial compounds such as TiC, $ZrH_{1.92}$, Mn_4N etc .
- Transition elements have a tendency to form coordination compounds with a species that has an ability to donate an electron pair to form a coordinate covalent bond.
- In the inner transition elements there are two series of elements. 1) Lanthanoids (previously called lanthanides) 2) Actinoids (previously called actinides)
- Lanthanoids have general electronic configuration [Xe] $4f^{1-14}$ $5d^{0-1}$ $6s^2$
- The common oxidation state of lanthanoides is +3
- As we move across 4f series, the atomic and ionic radii of lanthanoids show gradual decrease with increse in atomic number. This decrese in ionic size is called lanthanoid contraction.
- The electronic configuration of actinoids is not definite. The general valence shell electronic configuration of 5f elements is represented as [Rn]$5f^{0-14}6d^{0-2}7s^2$.
- Like lanthanoids, the most common state of actinoids is +3. In addition to that actinoids show variable oxidation states such as +2 , +3 , +4 ,+5,+6 and +7.

EVALUATION

CFF541

Choose the best answer:

1. Sc(Z=21) is a transition element but Zinc (z=30) is not because

   a) both Sc^{3+} and Zn^{2+} ions are colourless and form white compounds.

   b) in case of Sc, 3d orbital are partially filled but in Zn these are completely filled

   c) last electron as assumed to be added to 4s level in case of zinc

   d) both Sc and Zn do not exhibit variable oxidation states

2. Which of the following d block element has half filled penultimate d sub shell as well as half filled valence sub shell?

   a) Cr b) Pd

   c) Pt d) none of these

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